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Unit 10 – Database Development



Database in the Police and Bank

This presentation explains how database work in the police and banking sectors.

Overview of the Police Database.

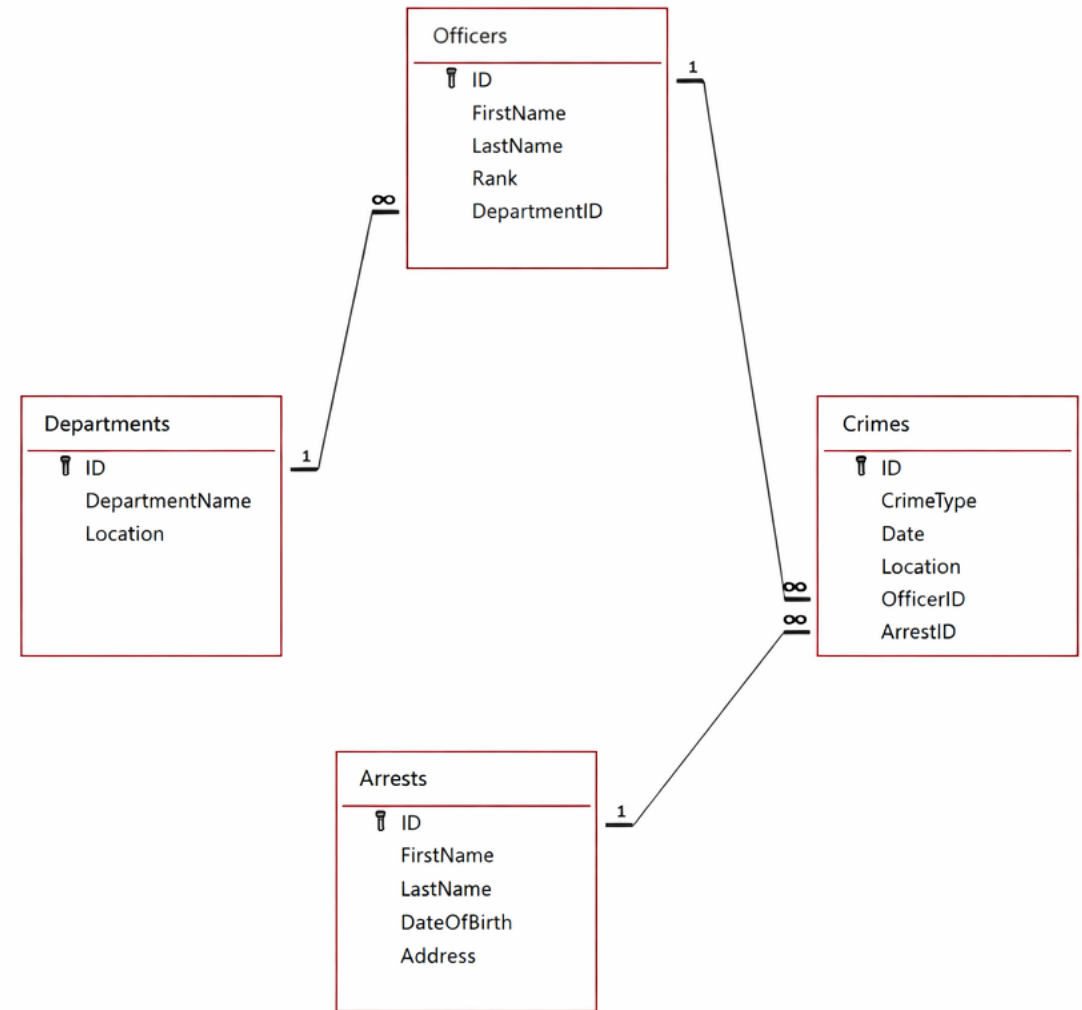


Police sector databases are used to store and manage information related to law enforcement. They include records such as officers, departments, crimes and offenders (see diagram aside).

The database stores important and sensitive data that helps police staff carry out their work efficiently. This includes officer details, crime, offender records, arrest information and case statuses.

Using a police database improves organisation, speeds up investigations, and helps ensure public safety by providing quick access to accurate information.

This ERD shows how tables are linked using primary and foreign keys. One-to-many relationships reduce data duplication and improve data integrity.



Tools and Techniques Used in the Police Database

- **Tables** are used to store data in a structured way such as officers crimes offenders and arrests with each table storing one type of data
- **Field characteristics (data types)** are used to ensure the correct type of data is entered

Officer ID uses a **Number (Max char: 50)** data type

Name fields use **Text (Max char: 10)**

Date reported uses **Date/Time**

Postcode uses **Short Text (Max char: 8)**

- **Validation rules** are used to reduce errors for example a postcode must follow the correct UK format and date fields cannot accept future dates
- **Relationships** link tables together for example officers are linked to crimes using an officer ID This allows one officer to be connected to many crimes and prevents duplicate data being stored
- **Queries** are used to search for specific information such as finding all crimes linked to one offender or all cases handled by one officer

What are the benefits of the Police sector for the database?

All police data is stored in one place – this allows officers and staff to quickly access information about crimes, offenders, and cases without searching through paper files.

THERE ARE MANY BENEFITS OF A POLICE SECTOR DATABASE SOME OF THESE ARE AS FOLLOWS:

What are the benefits of the Police sector for the database?

There are many BENEFITS OF A Police sector database some of these are as follows:

All police data is stored in one place – this allows officers and staff to quickly access information about crimes, offenders, and cases without searching through paper files.

Centralised records help link crimes to offenders and officers, making investigations faster and more efficient.

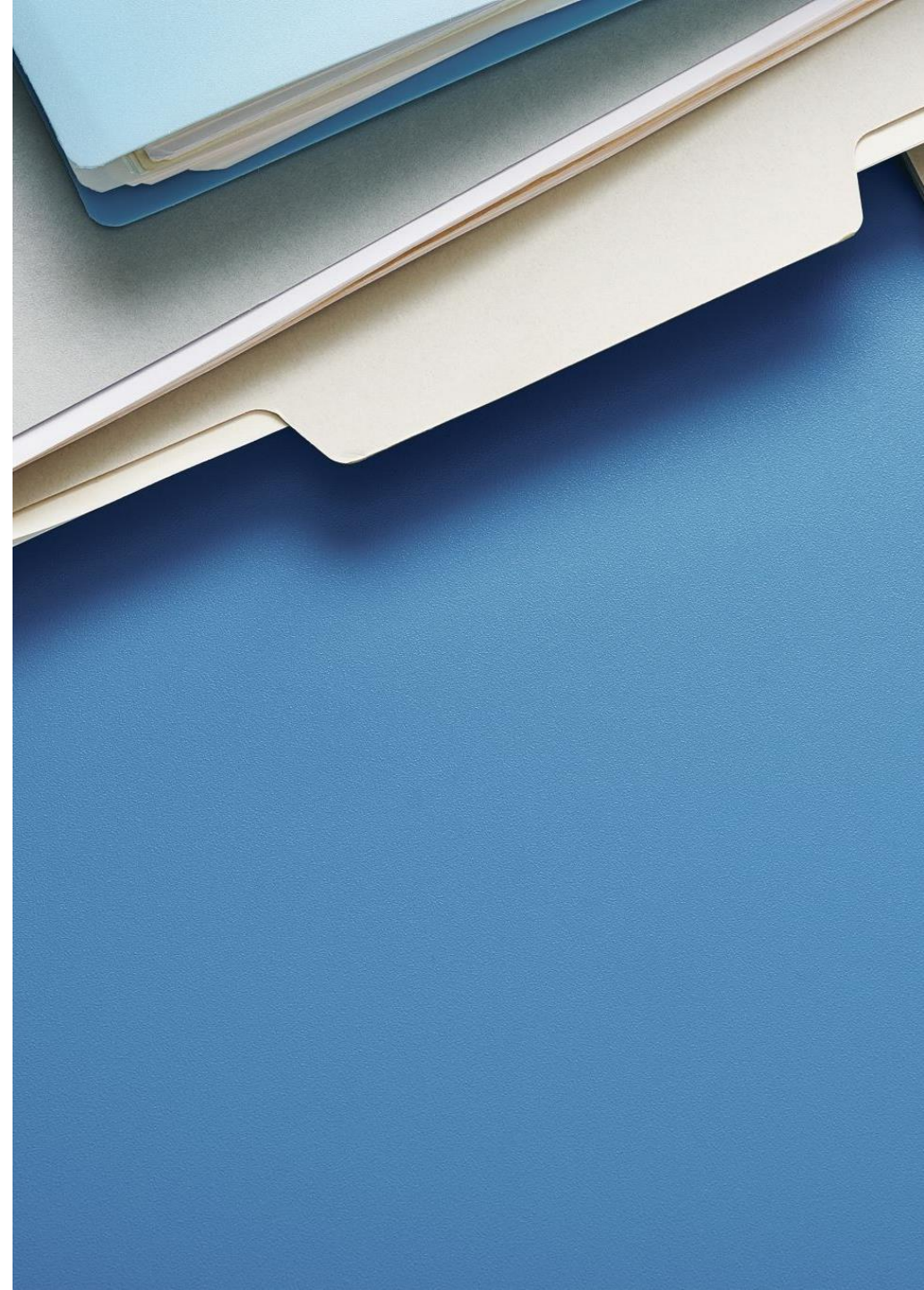
The database improves accuracy and reduces errors by keeping information up to date and consistent.

Sensitive information is securely stored and only accessible to authorised police staff.

Using a police database saves time, improves organisation, and supports better decision-making in law enforcement.

What Would the Police USE the Database for?

- All police data is stored in one place which allows officers and police staff to easily access information about crimes offenders and cases without searching through paper files
- Centralised records help link crimes to offenders and officers, making investigations faster and more efficient.
- The database improves accuracy and reduces errors by keeping information up to date and consistent
- Sensitive information is securely stored and only accessible to authorised police staff.
- Using a police database saves time, improves organisation, and supports better decision-making in law enforcement.



Strengths

1

The data can be accessed from different police stations and departments which allows officers to view crime reports offender records and case information without needing to recreate files

2

Centralised data improves organisation and allows investigations to be carried out faster and more efficiently

3

Sensitive police information is securely stored and only accessible to authorised staff

Weaknesses

1

Police databases contain highly sensitive information which makes them a target for cyber attacks if security is not properly maintained

2

System failures or downtime can delay police work and slow down investigations

3

Police staff require training to use the database correctly which can take time and additional cost

Conclusion for police

- The police sector database plays an important role in modern law enforcement. It allows police forces to store and manage large amounts of sensitive data in one central system. This improves organisation, accuracy, and efficiency when handling crimes, offenders, and investigations. Overall, the use of a police database supports faster investigations, better decision-making, and improved public safety.

Overview of the Bank Database

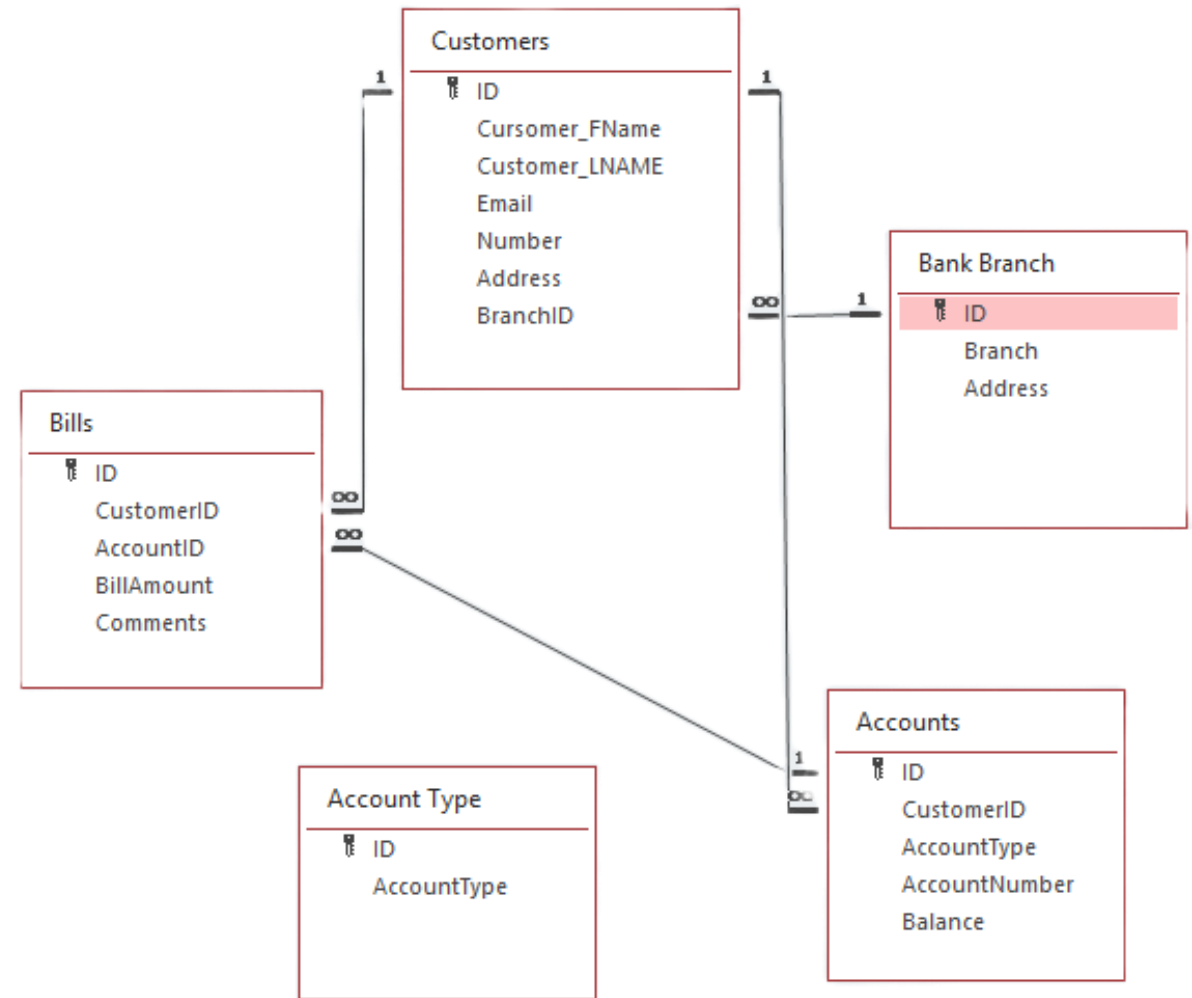


Bank sector databases are used to store and manage financial information related to customers and banking services. They include records such as customers, accounts, transactions, loans, and branches

The database stores important and sensitive financial data that helps bank staff carry out their work efficiently. This includes customer details, account balances, transaction history, loan information, and payment records

Using a bank database improves organisation, speeds up financial operations, and helps ensure security and accuracy by providing quick access to reliable and up-to-date information

This ERD shows how bank database tables are linked using primary and foreign keys. One-to-many relationships reduce data duplication and improve data integrity.



Tools and Techniques Used in the Bank Database

- **Tables** store customer account transaction and loan data in separate structured tables
- **Field characteristics (data types)** ensure accurate financial data entry

Customer ID uses **Number**

Customer name uses **Short text (Max char: 50)**

Account balance uses **Currency**

Sort code **Short Text(Max char:6)**

Account number use **Short Text (Max char: 8)**

Postcode uses **Short Text(Max char: 8)** because it contains letters and numbers

- **Validation rules** prevent incorrect data for example account numbers must be a fixed length and balances cannot be negative
- **Relationships** link customers to their accounts and transactions using a customer ID This allows one customer to have multiple accounts while keeping customer details stored only once
- **Queries** allow staff to quickly find customer details transaction history or loan information

Account number and sort code use Short Text instead of Number because they can start with zero.

Using Number data type would remove leading zeros and cause incorrect account details to be stored.



What are the benefits of the Bank sector for the database?

All bank data is stored in one place which allows bank staff to quickly access information about customers accounts transactions and loans without searching through paper files Centralised records help manage customer accounts process payments faster and keep financial information accurate and up to date The database also improves security by protecting sensitive financial data and supports better decision making within the bank

What are the benefits of the Bank sector for the database?



All bank data is stored in one place – this allows bank staff to quickly access information about customers accounts transactions and loans without searching through paper files.



Centralised records help link customers to their accounts and transactions making banking operations faster and more efficient.



The database improves accuracy and reduces errors by keeping financial information up to date and consistent.



Sensitive financial information is securely stored and only accessible to authorised bank staff.



Using a bank database saves time improves organisation and supports better decision making within the bank.

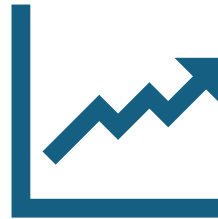
What Would the Bank USE the Database for?

-
- All bank data is stored in one place which allows bank staff to easily access information about customers accounts transactions and loans without searching through paper files
 - Centralised records help link customers to their accounts payments and loans making banking processes faster and more efficient
 - The database improves accuracy and reduces errors by keeping financial information up to date and consistent
 - Sensitive financial information is securely stored and only accessible to authorised bank staff
 - Using a bank database saves time improves organisation and supports better decision making within the bank

Strengths



The data can be accessed from different bank branches and departments which allows staff to view customer records account details transaction history and loan information without needing to recreate files



Centralised data improves organisation and allows banking processes to be completed faster and more efficiently

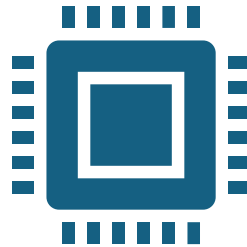


The database improves accuracy by keeping financial information up to date and consistent

Weaknesses



Banking databases store highly sensitive financial information which means they can be a target for cyber attacks if security is not properly maintained



System failures or downtime can prevent staff and customers from accessing accounts and services



Databases require regular maintenance updates and staff training which can increase costs for the bank

Conclusion for bank

- The bank sector database plays an important role in modern banking. It allows banks to store and manage large amounts of sensitive financial data in one central system. This improves organisation, accuracy, and efficiency when handling customer accounts, transactions, loans, and payments. Overall, the use of a bank database supports faster services, better decision making, and increased security for both the bank and its customers.

Final Summary

This presentation has explored two different types of databases used in the police and banking sectors. Each database is designed for a different purpose but uses similar tools and techniques such as tables relationships validation rules and queries. The police database is used to manage crime offender and investigation data while the bank database manages customer accounts transactions and financial records. Overall databases improve productivity accuracy and usability but also have weaknesses such as security risks and system downtime.